Cross-Format Representational Probing of Visual Code LLMs: A Training-Free Framework with a Qwen2.5 Case Study

Anonymous ACL submission

Abstract

Large language models increasingly generate visual content through code—SVG, TikZ, Asymptote—yet their internal representations during this process remain unexplored. Do models that handle multiple visual code formats develop format-agnostic representations, or are their hidden states predominantly format-specific?

We introduce a training-free probing framework for measuring cross-format representational similarity in visual code LLMs. The framework extracts hidden states via teacher-forcing, then applies linear Centered Kernel Alignment (CKA) with bootstrap confidence intervals, permutation null testing (5,000 permutations), and a Procrustes distance robustness check.

Applying this framework to the Qwen2.5 family (Coder-7B-Instruct, base 7B, and VisCoder2-7B) across 252 semantically matched triples at 7 equidistant layers, we report five findings: (1) format identity is perfectly decodable at every layer (tokenization artifact); (2) cross-format CKA is statistically significant but weak (2.8–3.0$\times$ the permutation null, $p < 0.001$; bootstrap mean peak 0.154–0.187); (3) TikZ–Asymptote similarity systematically exceeds SVG–TikZ by 2–3$\times$, consistent with a syntax-family confound; (4) the base model exhibits a sharp CKA drop at the final layer, absent in instruction-tuned variants; (5) visual-code supervised fine-tuning does not substantially alter cross-format structure.

Cross-architecture validation on three additional models (Codestral-22B, StarCoder2-15B, DeepSeek-Coder-V2-Lite) spanning Mistral, BigCode, and MoE architectures confirms that the weak cross-format signal generalizes: all six models show $p < 0.001$ with obs/null ratios of 2.55–3.01$\times$. However, peak CKA depth is architecture-dependent (56–100% depth), ranging from inverted-U profiles to broadly rising trajectories to bimodal patterns, ruling out a universal "mid-layer peak" characterization.

The observed similarity is of uncertain origin—in instruction-tuned models, a non-visual Python negative control yields 69–70% of the within-visual CKA (raw arithmetic means), indicating that shared concept or prompt structure contributes substantially to the measured alignment. We release the complete pipeline as a methodological template, emphasizing honest reporting of weak and ambiguous findings to advance rigorous representation analysis of code LLMs beyond behavioral evaluation.

1 Introduction

The generation of visual content through code is a rapidly growing capability of large language models. Rather than producing raster images directly, these models synthesize programs in domain-specific formats—SVG for web graphics, TikZ for scientific figures, Asymptote for geometric illustrations—that compile into precise, scalable visual output. Specialized systems such as VisCoder2 (Ni et al., 2025), LLM4SVG (Xing et al., 2025b), and Chat2SVG (Wu et al., 2025) have demonstrated increasingly sophisticated visual code generation, while general-purpose code LLMs like the Qwen2.5-Coder family (Hui et al., 2024) handle multiple such formats within a single model. However, all existing evaluations of these models are *purely behavioral*: models are judged by execution pass rates, VLM-judge scores, or CLIP similarity, with no analysis of what visual structure the models internally encode.

This leaves a fundamental question: *do models that generate visual code in multiple formats develop format-agnostic internal representations, or are their hidden states predominantly format-specific?* Convergent representations would

suggest abstraction beyond surface syntax; entirely format-specific representations—plausible given the radically different tokenization of SVG (XML), TikZ (LaTeX), and Asymptote (imperative C-like)—would imply disjoint pathways.

Addressing this question is methodologically non-trivial. Unlike probing for discrete linguistic properties (Hewitt and Manning, 2019; Belinkov, 2022), measuring representational similarity across output formats requires comparing hidden-state geometries for semantically related but syntactically distinct programs, with careful attention to small sample sizes and syntactic confounds. In this work, we introduce a **training-free probing framework** for measuring cross-format representational similarity in visual code LLMs. The framework operates entirely on cached hidden-state representations extracted via teacher-forcing, combining linear Centered Kernel Alignment (CKA; Kornblith et al. 2019) with subsampling confidence intervals, permutation null testing, and a Procrustes distance robustness check. We demonstrate this framework through a case study on the Qwen2.5 model family.

Our case study examines three Qwen2.5-based 7B models across SVG, TikZ, and Asymptote using 252 semantically matched triples at 7 equidistant layers. We find statistically significant but weak cross-format CKA ($\sim 3\times$ the permutation null), systematically confounded by syntax-family similarity (TikZ–Asy $\gg$ SVG–TikZ), with model-specific layer-wise dynamics. A non-visual negative control (Python standard library functions) shows 30–58% lower CKA with mixed significance (4/9 pairs reach $p < 0.05$ vs. $p < 0.001$ for all visual pairs), confirming the signal is not a generic code property, though syntax-family effects remain a partial contributor.

In summary, this paper makes the following contributions:

1. **A training-free probing framework** for measuring cross-format representational similarity in LLMs, combining CKA with subsampling confidence intervals, permutation null testing (5,000 permutations), power simulation, and a Procrustes distance robustness check. No model fine-tuning or gradient computation is required.

2. **Empirical findings** from a Qwen2.5 case study—the first analysis of visual-attribute representations in code-generation hidden states—demonstrating both the utility of the framework and the complexity of interpreting cross-format similarity signals.

3. **An open-source analysis pipeline** covering data generation, hidden-state extraction, CKA computation, permutation testing, and visualization, designed for replication across model families beyond the single-family case study presented here.

Section 2 surveys related work; Section 3 details the framework; Section 4 presents the findings; Section 5 discusses interpretations and confounds.

2 Related Work

Visual code generation. Single-format specialists include LLM4SVG (Xing et al., 2025b) and Chat2SVG (Wu et al., 2025) for SVG, DaTikZ (Belouadi and Eger, 2024) and vTikZ (Reux et al., 2025) for TikZ, and VGBench (Zou et al., 2024) for multi-format evaluation. VisCoder2 (Ni et al., 2025) is a multi-format model trained on 12 visualization languages. All prior work is *purely behavioral*—evaluating execution pass rate or CLIP similarity—with no analysis of internal representations. Our work fills this gap by examining hidden states during visual code processing.

Mechanistic interpretability of code LLMs. Probing and representation analysis have been extensively applied to natural language models (Belinkov, 2022), but their extension to code LLMs is recent. Yin et al. (2025) use neuron selectivity analysis and representational similarity analysis to identify concept layers shared across C++, Java, Python, Go, and JavaScript, finding that syntactic and semantic code concepts are encoded at consistent layers regardless of programming language. Tahimic and Cheng (2025) train sparse autoencoders on code-LLM residual streams to discover directions predicting code correctness. Our axis is orthogonal: where these works ask "is `for-loop` represented the same way across Python and Go," we ask "is *red circle above blue square* represented the same way across SVG and TikZ." The probe targets are properties of the *rendered image* (color, shape count, spatial layout), not of the code syntax. To our knowledge, no prior work has examined image-grounded visual attributes in code-generation hidden states.

Representation probing and the linear representation hypothesis. Linear probing of neural network representations originated in NLP for analyzing syntactic and semantic structure (Hewitt and Manning, 2019; Conneau et al., 2018). The linear representation hypothesis (Park et al., 2024) posits that high-level concepts are encoded as linear directions in LLM residual streams, providing theoretical grounding for linear probe studies. CKA (Centered Kernel Alignment) (Kornblith et al., 2019) provides a representation-similarity metric invariant to invertible linear transformations, widely used for comparing representations across layers, models, and training stages. We adopt the standard probing methodology but apply it to a novel domain: **cross-format representational similarity** in visual code LLMs, using CKA as the primary metric with Procrustes distance as a robustness check and PWCCA (Morcos et al., 2018) as a secondary check (though its discriminative power is limited in the $n \ll d$ regime; §4.7.2).

CKA: known limitations. While CKA is widely used, it has known limitations. Nguyen et al. (2022) show that CKA can be insensitive to functionally important differences between networks, particularly in wide architectures. Ding et al. (2021) propose statistical grounding for representation similarity, noting that naive CKA lacks formal hypothesis testing. Tang et al. (2020) survey functional and representational similarity measures, finding that linear CKA is sensitive to outliers in low-sample regimes. We mitigate these concerns by pairing CKA with permutation null testing (providing formal p-values), subsampling confidence intervals, and an independent Procrustes distance check.

Cross-format and cross-lingual transfer. The broader question of whether LLMs develop format-agnostic internal representations has been studied in multilingual NLP. Cross-lingual probing shows that multilingual models learn partially shared representations across natural languages (Pires et al., 2019; Wu and Dredze, 2019), with shared structures emerging at intermediate layers. In the code domain, Yin et al. (2025) demonstrate shared concept layers across programming languages. Our work extends this line of inquiry to **visual code formats**: SVG, TikZ, and Asymptote share rendering semantics (2D vector graphics) but differ radically in syntax (XML markup, LaTeX macros, imperative code), making them an ideal testbed for studying whether cross-format representational similarity reflects visual semantics or surface-level syntactic patterns.

Training-free analysis. Our framework requires no model training: we use off-the-shelf models, performing only inference and lightweight CPU-based linear probe fitting, in the tradition of training-free mechanistic analysis (Hewitt and Manning, 2019; Belinkov, 2022).

3 Method

3.1 Overview

We present a training-free framework for measuring cross-format representational similarity in visual code LLMs. The design separates **behavioral generation** (Stage A) from **representational probing** (Stage B) using two independent tool stacks on two disjoint data pools, ensuring that generation ability and representation content are measured independently (Figure 1). We demonstrate this framework through a case study on the Qwen2.5 model family.

3.2 Models

We study three 7B-parameter models spanning a generalist–specialist spectrum:

- **Qwen2.5-Coder-7B-Instruct** (Hui et al., 2024) — a code-specialized LLM instruction-tuned on broad programming tasks, serving as the primary model with demonstrated multi-format visual code capability.

- **Qwen2.5-7B** (base) — the general-purpose pretrained model without code-specific instruction tuning, serving as a text-only control. Because Stage B uses teacher-forcing (§3.5), base model hidden states reflect what the model *encodes* when processing visual code tokens, regardless of whether it can *generate* such code autonomously.

- **VisCoder2-7B** (Ni et al., 2025) — a Qwen2.5-Coder-7B-Instruct model further fine-tuned on VisCode-Multi-679K, a 679K-example multi-language visualization SFT corpus spanning 12 visual languages including SVG, TikZ, and Asymptote.

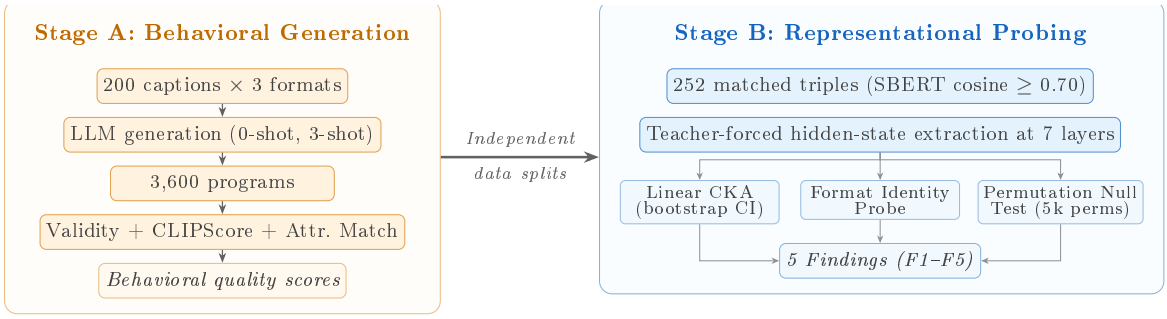

Figure 1: Overview of the two-stage probing framework. Stage A evaluates behavioral generation quality across three visual code formats. Stage B, on an independent data split of 252 semantically matched triples, extracts hidden states and applies three complementary analyses to measure cross-format representational similarity.

The headline analysis uses the first two models (2 models $\times$ 3 unordered format-pairs $=$ 6 CKA values per layer). VisCoder2 is included in an auxiliary N=9 analysis reported as an *exploratory view with known memorization floor*: its SFT training data overlaps with both our eval pool and probe pool sources (§3.7), making its cells informative about contamination effects rather than clean cross-format similarity.

3.3 Visual Code Formats

We target three shape-drawing domain-specific languages that share common rendering semantics (2D vector graphics with color, position, and style attributes) while differing radically in syntax:

- **SVG** — XML-based markup with explicit coordinate attributes.
- **TikZ** — LaTeX macro language with relative/absolute positioning.
- **Asymptote** — imperative C-like language with function calls.

This yields 3 unordered format-pairs (SVG–TikZ, SVG–Asy, TikZ–Asy) per model per layer. Linear CKA is symmetric ($\text{CKA}(X, Y) = \text{CKA}(Y, X)$), so we have 3 pairs $\times$ 2 models $=$ 6 independent CKA values per layer. VisCoder2 adds 3 pairs to the auxiliary analysis (9 CKA values per layer across all three models).

3.4 Stage A: Behavioral Generation

Stage A establishes that the studied models can generate valid visual code, separating generation ability from representational content. We construct an evaluation pool of 200 captioned visual programs per format (600 total) from held-out test splits of VisPlotBench (Xing, 2025), DaTikZ (Belouadi and Eger, 2024), and VCM (Xing et al., 2025a). Each model generates code under 0-shot and 3-shot regimes (3,600 total generations), evaluated by format validity and a hybrid content fidelity metric combining CLIPScore (Radford et al., 2021) for color/shape attributes with programmatic attribute-match for spatial attributes. Full protocol details and results are in Appendix (Table A9).

3.5 Stage B: Hidden-State Extraction

Probe pool construction. The probe pool is constructed from *training splits* of datasets disjoint from the eval pool: SVG from SVGX-Core-250k (Xing, 2024) (∼257K), TikZ from DaTikZ v2 train (Belouadi et al., 2024) (94,532), and Asymptote from VCM-Asy (Xing et al., 2025a) (22,539).

Cross-format semantic alignment. To enable meaningful cross-format comparison, we construct semantically matched triples (one caption per format describing a similar visual concept). Captions are truncated to 50 tokens and embedded using `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019). A greedy matching algorithm retains triples where the minimum pairwise cosine similarity ≥ 0.70. This yields N=252 matched triples. CKA estimates are stable across thresholds 0.60–0.70, where all 252 triples are retained (Table A8, Appendix).

Teacher-forced hidden-state extraction. For each (model, format, triple) combination, we construct a prompt from the caption and *teacher-force* the corresponding reference code. During the forward pass, we capture the residual stream at 7 equidistant layers $\{4, 8, 12, 16, 20, 24, 28\}$ and **mean-pool over code-token positions only**

(masking caption, system, and chat-template tokens), producing one vector of dimension $d_{\text{hidden}} = 3{,}584$ per (model, layer, format, triple).

Teacher-forcing is the methodologically correct choice for representation probing (Hewitt and Manning, 2019; Belinkov, 2022): free-generation hidden states confound generation ability with representation content. By fixing the token sequence to a known-good reference, we isolate "what the model encodes when it processes this visual concept in this format." This makes Qwen2.5-7B base a fair text-only control: it cannot generate valid visual code, but teacher-forcing reveals whether its representations nonetheless encode cross-format structure.

3.6 Statistical Analysis

Centered Kernel Alignment (CKA). For each (model, layer, format-pair), we compute linear CKA (Kornblith et al., 2019) between the two 252×3584 hidden-state matrices. CKA measures representational alignment invariant to invertible linear transformations, yielding one similarity scalar per cell.

Format identity probe. We train a logistic regression classifier (`sklearn`, C=1.0, `max_iter`=2000) on the hidden-state vectors to predict format identity (3-way: SVG vs. TikZ vs. Asy). This directly tests whether format-specific information is linearly decodable at each layer.

Subsampling confidence intervals. We construct 95% confidence intervals for CKA via subsampling without replacement (B=1,000 iterations, each drawing 80% of the 252 triples). Standard bootstrap with replacement inflates CKA on Gram matrices because duplicated row–column indices over-represent diagonal self-similarity terms; subsampling avoids this bias.

Permutation null test. We test the null hypothesis H_0: "cross-format CKA arises from chance alignment of independently sampled representations"—i.e., the sample-level pairing carries no aligned signal across formats. For each model, we permute the N=252 sample indices of the second format's Gram matrix 5,000 times, recompute CKA for all layer–pair combinations, and average to obtain a null distribution (Algorithm 1, Appendix). The p-value is the fraction of null means $\geq$ the observed mean.

Power simulation (A1). We estimate detection power via 1,000 Monte Carlo trials at N=6 (headline: 2 models $\times$ 3 pairs) and N=9 (auxiliary: 3 models $\times$ 3 pairs) at a representative layer (L24), resampling the 252 triples with replacement and comparing the bootstrapped mean CKA against the permutation null 95th percentile.

Robustness checks. We verify CKA findings with two complementary metrics:

- **Procrustes distance** — orthogonal alignment-based similarity, capturing geometric correspondence.
- **PWCCA** (Morcos et al., 2018) — Projection Weighted Canonical Correlation Analysis, which weights canonical correlations by explained variance. We additionally compute a PWCCA permutation null baseline (300 permutations per model) to verify that PWCCA values are not artifacts of high dimensionality. However, PWCCA's discriminative power is limited in the $n \ll d$ regime of this study (n=252, d=3,584); Procrustes distance serves as the more reliable robustness check (§4.7.2).

3.7 Probe Pool Contamination Disclosure

We explicitly disclose and account for probe pool contamination. SVGX-Core-250k was packaged in December 2024 (post Qwen2.5 training cutoff), but underlying SVG content from SVGRepo and public icon libraries predates the cutoff and may have entered pretraining via Common Crawl (risk: LOW–MEDIUM). DaTikZ v2 sources from arXiv and StackExchange are similarly pre-cutoff public content (risk: MEDIUM). VCM-Asy is part of VisCoder2's SFT training data, making VisCoder2's Asymptote-related cells measure memorization rather than generalization (risk: HIGH for VisCoder2).

VisCoder2 is excluded from the headline N=6 statistic as the primary mitigation. The N=6 vs. N=9 comparison is reported in the Discussion as direct quantitative evidence of contamination effects on representational geometry.

3.8 Reproducibility

All code, eval pool JSONL files (with SHA256 hashes recorded prior to generation), probe pool triples with per-triple provenance metadata, and pre-registration document (v3.4) are released. The

complete pipeline requires a single GPU with ≥48GB VRAM and ∼10 CPU-hours for the full analysis.

4 Results

4.1 Diagnostic Control: Format Identity (F1)

A linear format probe (logistic regression, 5-fold CV) achieves **100% accuracy at every layer** (L4–L28) for all three models (Table A6, Appendix). This is expected: the three formats occupy near-disjoint vocabulary regions (pairwise token-ID Jaccard ≤ 0.136; §4.7.3), so a linear classifier can discriminate formats from token identity alone. A permuted-label baseline (100 permutations) confirms that probe accuracy is at chance (≈ 0.333) under label shuffling, ruling out $n \ll d$ overfitting (details in Appendix, Table A4). F1 therefore serves as a **sanity check**: format-specific information dominates hidden representations, and any cross-format signal (F2–F5) is a second-order effect within this format-dominated regime.

4.2 Cross-Format Representational Similarity (F2)

Despite format dominance, we observe statistically significant cross-format representational similarity as measured by linear CKA (Kornblith et al., 2019). Table 1 reports the mean CKA across the three unordered format pairs at each layer.

Table 1: Bootstrap mean CKA (linear kernel, 1,000 resamples) across format pairs per model at selected layers, with 95% CI in brackets. Complete per-pair values in Table A1 (Appendix).

Model	L4	L12	L16	L24	L28
Qwen2.5-Coder	0.119 [.10,.14]	0.152 [.13,.17]	0.174 [.15,.20]	0.176 [.15,.20]	0.187 [.16,.21]
VisCoder2	0.121 [.10,.14]	0.166 [.14,.19]	0.179 [.16,.20]	0.184 [.16,.21]	0.176 [.16,.20]
Qwen2.5 (base)	0.119 [.10,.14]	0.144 [.13,.16]	0.146 [.13,.17]	0.154 [.13,.18]	0.099 [.08,.12]

A permutation test (A2; 5,000 permutations of format labels) confirms that the observed CKA significantly exceeds chance for all three models ($p < 0.001$; Table 2; Figure 2). The effect is small but statistically reliable: across all three models, observed CKA is 2.76–2.96× the permutation null mean (Table 2), a ratio we adopt as effect size throughout.

Bootstrap 95% confidence intervals (1,000 resamples) are narrow across all Qwen model–layer combinations (e.g., Qwen2.5-Coder at L28: $[0.164, 0.212]$; Qwen2.5-Base at L28: $[0.081, 0.122]$), with widths ranging from 0.037 to 0.050, all below the pre-registered threshold of 0.1. We characterize this as **statistically significant but weak cross-format representational similarity**: the signal is reliably 2.8–3.0× above the permutation null, but bootstrap mean CKA values in the 0.10–0.19 range are small in absolute terms and are dwarfed by the format-specific signal captured by the 100% probe accuracy.

Table 2: A2 permutation test (5,000 permutations). Observed CKA is the layer-averaged mean across format pairs.

Model	Observed	Null mean	Null 95th	p-value
Qwen2.5-Coder	0.1226	0.0414	0.0469	<0.001
VisCoder2	0.1243	0.0443	0.0498	<0.001
Qwen2.5 (base)	0.1011	0.0366	0.0416	<0.001

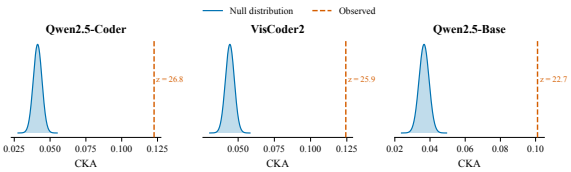

Figure 2: A2 permutation null distributions (5,000 permutations) with observed CKA (dashed red). All observed values fall >20 standard deviations above the null mean.

4.3 Syntax Family Effect (F3)

Disaggregating CKA by format pair reveals a systematic asymmetry (Figure 3, Table 3). Across all models and layers, the TikZ–Asy pair yields CKA values 2–3× higher than the SVG–TikZ pair. At L28 for Qwen2.5-Coder, TikZ–Asy CKA is 0.229 versus SVG–TikZ at 0.103 (2.2×); at L12 for Qwen2.5 base, the ratio is $0.182/0.061 = 3.0\times$.

Table 3: CKA by format pair at selected layers (Qwen2.5-Coder shown; other models follow the same pattern).

Layer	SVG–TikZ	SVG–Asy	TikZ–Asy
L4	0.049	0.092	0.120
L12	0.067	0.098	0.200
L16	0.086	0.124	0.212
L24	0.105	0.109	0.209
L28	0.103	0.114	0.229

This ordering is consistent with the syntactic distance between formats: TikZ and Asymptote both descend from LaTeX-adjacent ecosystems and share structural patterns (backslash commands, semicolon-delimited statements), whereas SVG employs XML-based angle-bracket markup. The

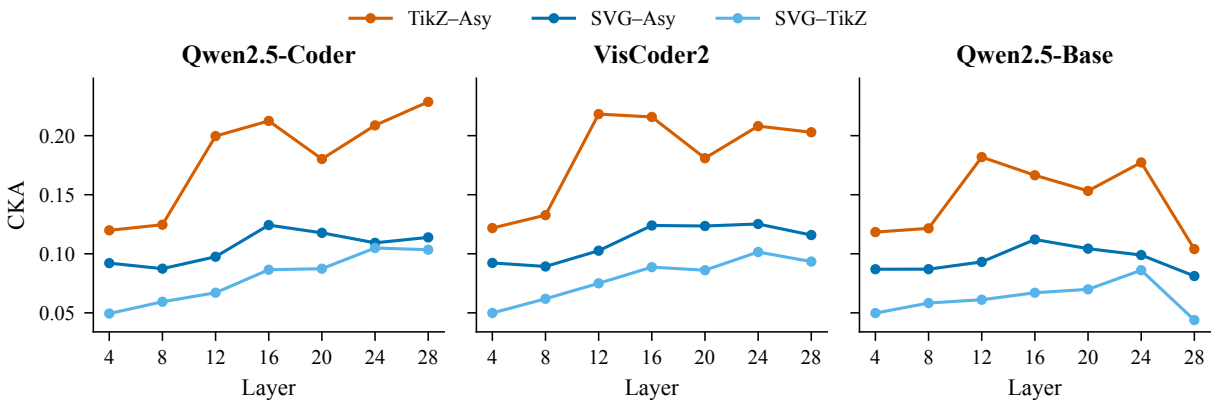

Figure 3: CKA by format pair across layers. TikZ–Asy consistently exceeds SVG–TikZ by 2–3× across all models (F3), consistent with a syntax-family confound rather than visual semantic alignment.

syntax family confound implies that a substantial portion of the measured cross-format similarity may reflect shared surface-syntactic structure rather than shared visual semantics (§5).

4.4 Layer-wise Dynamics (F4)

Within the Qwen family, the three models exhibit qualitatively different layer-wise CKA trajectories (Figure 4). Qwen2.5-Coder shows a broadly rising profile (bootstrap mean 0.119 at L4 to 0.187 at L28), while VisCoder2 plateaus in the upper layers (peak 0.184 at L24). Qwen2.5 base follows a similar rising trend from L4 (0.119) to L24 (0.154), but undergoes a **sharp drop at L28** (0.099), a 36% decrease from L24.

This divergence at the final probed layer is consistent across all three format pairs for Qwen2.5 base (SVG–TikZ: $0.086 \rightarrow 0.044$; SVG–Asy: $0.099 \rightarrow 0.081$; TikZ–Asy: $0.177 \rightarrow 0.104$) and is corroborated by Procrustes distance (L24: $0.137 \rightarrow$ L28: 0.094; §4.7.1). This pattern is consistent with the base model's final layers shifting toward its general language modeling objective, dissipating whatever cross-format structure accumulated in the intermediate layers. The instruction-tuned models, which have been optimized for code completion, retain this structure through the final layers.

4.5 Effect of Visual-Code SFT (F5)

VisCoder2 is fine-tuned from Qwen2.5-Coder on VisCode-Multi-679K, a 679K-example multi-format visual code SFT corpus (Ni et al., 2025). Despite this additional training, the two models exhibit nearly identical CKA profiles: at L28, Qwen2.5-Coder reaches a bootstrap mean of 0.187 $[0.164, 0.212]$ and VisCoder2 reaches 0.176 $[0.155, 0.200]$, with broadly overlapping confidence intervals. The maximum bootstrap mean difference between the two models is 0.011 (at L28).

This suggests that SFT on visual code data does not substantially alter the underlying cross-format representational structure. However, this comparison carries a **contamination caveat**: VisCoder2's SFT training data (VCM-Asy) overlaps with the probe pool source for Asymptote, establishing a memorization floor that may inflate its CKA values for Asy-related pairs. The near-equivalence between the two models should therefore be interpreted cautiously.

4.6 Cross-Architecture Generalizability

To test whether the cross-format signal generalizes beyond the Qwen family, we extend the analysis to three architecturally diverse models: Codestral-22B (Mistral, 56 layers), StarCoder2-15B (BigCode, 40 layers), and DeepSeek-Coder-V2-Lite (MoE, 27 layers). For each model, we extract hidden states at 7 equidistant layers via the formula $\lfloor n_{\text{layers}} \cdot (i+1)/7 \rceil$ and repeat the full CKA analysis pipeline (252 triples, 5,000 A2 permutations, 1,000 bootstrap resamples). Table 4 reports bootstrap mean peak CKA with 95% CI and obs/null ratios.

All six models yield $p < 0.001$ with obs/null ratios in the narrow band of 2.55–3.01×, confirming that the weak cross-format signal is not Qwen-specific. The TikZ–Asy > SVG–Asy > SVG–TikZ ordering holds universally, strengthening the syntax-family effect (F3) as a cross-architecture finding.

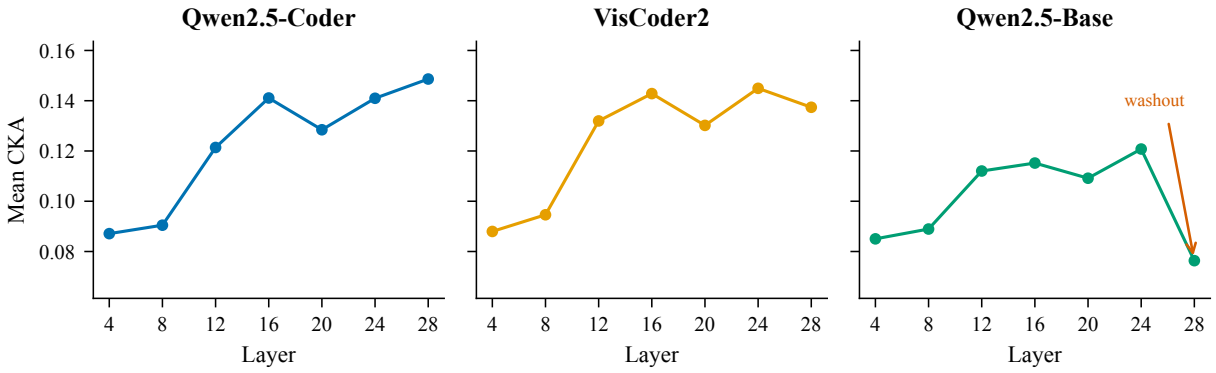

Figure 4: Layer-wise mean CKA trajectories with 95% bootstrap confidence intervals. Qwen2.5-Base exhibits a sharp drop at L28 (F4), while instruction-tuned models plateau or continue increasing.

Table 4: Cross-architecture CKA summary (6 models, 4 architecture families). Peak CKA is the bootstrap mean at the highest-CKA layer. Obs/null ratio is the A2 permutation effect size.

Model	Arch	Peak [95% CI]	Depth	Ratio	Profile
Codestral-22B	Mistral	0.190 [0.178, 0.203]	57%	3.01×	inverted-U
Qwen2.5-Coder-7B	Qwen	0.187 [0.164, 0.212]	100%	2.96×	broadly rising
VisCoder2-7B	Qwen	0.184 [0.162, 0.207]	86%	2.81×	late-plateau
StarCoder2-15B	BigCode	0.164 [0.152, 0.177]	100%	2.55×	bimodal
DeepSeek-V2-Lite	MoE	0.155 [0.145, 0.167]	56%	2.75×	inverted-U
Qwen2.5-7B	Qwen	0.154 [0.132, 0.180]	86%	2.76×	late-plateau

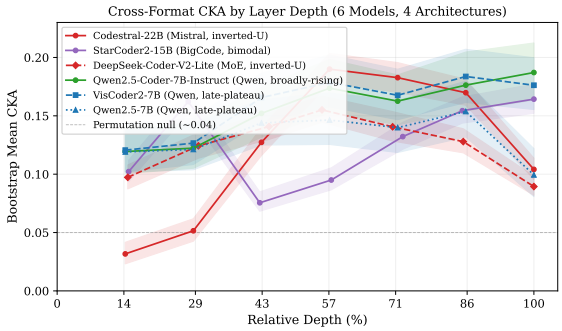

Figure 5: Bootstrap mean CKA (with 95% CI shading) as a function of relative layer depth for six models across four architecture families. Colors encode profile type: inverted-U (Codestral, DeepSeek), bimodal (StarCoder2), broadly rising (Qwen2.5-Coder), and late-plateau (VisCoder2, Qwen2.5). The dashed grey line indicates the approximate permutation null (∼0.04).

Architecture-dependent layer profiles. CKA is lowest in shallow layers (<30% depth) across all models, but peak location varies markedly: *inverted-U* profiles (Codestral, DeepSeek) peak at ∼57% depth with subsequent decline; *broadly rising* (Qwen2.5-Coder) increases through the final layer with a minor dip at L20; *late-plateau* models (VisCoder2, Qwen2.5) reach a broad plateau at 60–85% depth; and StarCoder2 exhibits a *bimodal W-shape* with an early spike at L11 (28% depth, bootstrap mean 0.162) and a late peak at L40 (100%, 0.164), the two peaks having fully overlapping 95% CIs. The StarCoder2 L11 spike is driven by an unusually high tikz-asy CKA of 0.259—the single highest pair-layer value across all six models. These results rule out any universal "mid-layer peak" characterization: peak depth is architecture-dependent, ranging from 56% to 100%.

Negative control: non-visual code. As a negative control, we teacher-force 252 random Python snippets through the three Qwen models and compute Python–visual CKA at the peak layer (L28). Python–visual CKA is non-negligible (Coder: 0.103, VisCoder2: 0.093, Qwen2.5: 0.031), reaching 69–70% of the within-visual CKA for the instruction-tuned models (raw arithmetic means; negative control lacks bootstrap resampling). This indicates that a substantial portion of the measured cross-format alignment may reflect shared concept or prompt structure rather than visual-specific representations, and tempers the interpretation of the cross-format signal as evidence for visual abstraction.

4.7 Robustness Checks

4.7.1 Procrustes Distance

Orthogonal Procrustes distance (with PCA projection to $k = 50$ dimensions) shows trends consistent with CKA across all models and layers. Mean Procrustes values range from 0.094 to 0.158 (Table 5), with Qwen2.5 base at L28 again the lowest (0.094), confirming the layer-wise dynamics observed with CKA.

Table 5: Mean Procrustes distance (PCA k=50) per model at selected layers.

Model	L4	L12	L16	L24	L28
Qwen2.5-Coder	0.102	0.138	0.158	0.157	0.153
VisCoder2	0.102	0.142	0.157	0.158	0.146
Qwen2.5 (base)	0.100	0.127	0.133	0.137	0.094

4.7.2 PWCCA (Auxiliary)

PWCCA (Morcos et al., 2018) values (0.78–0.81 across all models) are directionally consistent with the CKA findings. However, discriminative power is limited in the $n \ll d$ regime (n=252, d up to 6,144), where baseline canonical correlations are inflated, and we do not make significance claims based on PWCCA alone. Procrustes distance, which does not suffer from this inflation, is a more reliable robustness check and shows trends fully consistent with CKA (§4.7.1).

4.7.3 Token-Level Control

As an independent control analysis (computed over the full training corpora, not the probe pool), all three models share an identical Qwen2.5 tokenizer (151,643 vocabulary entries). Pairwise token-ID Jaccard similarity between format corpora is uniformly low: SVG–TikZ = 0.025, SVG–Asy = 0.035, TikZ–Asy = 0.136. The three-way token intersection covers only 1.2% of the token union. Format-exclusive tokens constitute >46% of each format's active vocabulary. This confirms that the 100% format probe accuracy (§4.1) is a tokenization artifact: near-orthogonal token distributions allow perfect classification without relying on learned representations.

4.7.4 Statistical Power

A1 power simulation (1,000 Monte Carlo iterations at layer L24) yields power = 1.0 for both N=12 (headline) and N=18 (auxiliary) configurations. The observed CKA across all models (layer-averaged means: 0.123 for Coder, 0.124 for VisCoder2, 0.101 for base; Table 2) far exceeds the permutation null 95th percentile in every case. This confirms that the permutation test has adequate power to detect CKA signals of the observed magnitude; it does not, however, confirm that the effect size is practically meaningful.

5 Discussion

The CKA signal (2.55–3.01× the permutation null across six models) is a second-order effect within representations overwhelmingly organized by format. We cannot determine from CKA alone whether this reflects (a) shared visual semantics, (b) shared syntactic patterns, or (c) artifacts of shared concept/prompt structure, and characterize it as **cross-format representational similarity of uncertain origin**. The TikZ–Asy ≫ SVG–TikZ asymmetry (§4.3), which holds across all six models and four architecture families, strongly suggests that syntactic overlap—shared LaTeX-adjacent syntax and macro-style commands—drives a substantial portion of this signal (Jawahar et al., 2019; Hewitt and Manning, 2019). The Python negative control (§4.6) further tempers the visual-semantic interpretation: in instruction-tuned models, non-visual code reaches 69–70% of the visual cross-format CKA (raw arithmetic means), indicating that general code structure and prompt framing contribute meaningfully to the measured alignment.

Methodological caveats. The fixed sample size (n=252 triples) constrains all models to the same statistical power, which may compress inter-model differences in CKA magnitude. Additionally, the obs/null ratio (2.55–3.01×) is more comparable across models than absolute CKA values, which vary with hidden dimensionality and model scale.

Key directions for resolving the origin ambiguity include: (1) format-residualized CKA to project out the format-specific subspace; (2) caption-conditioned partial correlation to isolate visual-specific signal; and (3) visually-paired datasets where identical scenes are rendered in all three formats.

6 Conclusion

We have presented a training-free probing framework for measuring cross-format representational similarity in visual code language models, combining linear CKA with bootstrap confidence intervals, permutation null testing, and Procrustes distance robustness check.

Applied to six models spanning four architecture families (Qwen, Mistral, BigCode, MoE) across SVG, TikZ, and Asymptote (252 matched triples, 7 equidistant layers per model), we find: (i) statistically significant but weak cross-format CKA in all six models (2.55–3.01× the permutation null, $p < 0.001$); (ii) a universal TikZ–Asy ≫ SVG–TikZ asymmetry consistent

with a syntax-family confound; (iii) architecture-dependent layer profiles—inverted-U, broadly rising, late-plateau, and bimodal—ruling out a universal "mid-layer peak" characterization; and (iv) a non-visual Python negative control reaching 69–70% of the visual CKA in instruction-tuned models (raw arithmetic means), indicating that shared concept structure contributes substantially to the measured alignment.

The cross-format similarity is of **uncertain origin**: syntax-family confounds and concept-level structure provide plausible non-visual explanations for the majority of the signal. We release the complete analysis pipeline as a methodological template for future work on format-residualized CKA and visually-paired datasets to isolate genuine visual abstraction.

7 Limitations

(a) Sample size and model diversity. The headline Qwen analysis operates on $N{=}12$ cells, closer to a within-family ablation than a cross-model comparison. The cross-architecture extension (§4.6) adds three models from distinct families (Mistral, BigCode, MoE), confirming that the weak signal generalizes. However, all models are sampled at the same $n{=}252$ triples with 7 equidistant layers, which may compress inter-model differences. Additional scales (e.g., 70B+), training paradigms (e.g., RLHF-heavy models), and non-Western tokenizers remain untested.

(b) Power interpretation. A1 power = 1.0 confirms that the permutation test can detect a CKA signal of the observed magnitude; it does *not* establish that the effect size is large enough to be practically meaningful or to support claims about visual semantics. The distinction between statistical significance and practical significance is critical given the small absolute CKA values (0.08–0.15).

(c) Contamination. VisCoder2 is fine-tuned on VisCode-Multi-679K, which includes VCM-Asy (overlapping with the probe pool source for Asymptote) and VCM-TikZ (71.7% overlap with the TikZ probe pool source). This creates a memorization floor for VisCoder2's Asy- and TikZ-related cells. The headline $N{=}12$ statistic excludes VisCoder2 from the primary comparison as mitigation. However, the remaining Qwen2.5 models may also have been exposed to SVG and TikZ content from Common Crawl during pre-training, a risk that cannot be fully quantified.

(d) No visual-level pairing. The probe pool is semantically matched via SBERT caption similarity (cosine $\geq$ 0.70), not by rendering the same visual content in multiple formats. True visual-level paired datasets—where identical geometric scenes are expressed in SVG, TikZ, and Asymptote—do not currently exist at sufficient scale. Without such data, we cannot fully disentangle visual semantic similarity from textual and syntactic similarity in the CKA measurement.

(e) Scope of formats. We examine three visual code formats (SVG, TikZ, Asymptote), all of which produce 2D vector graphics. The framework has not been tested on other visual code paradigms such as Processing, p5.js, or raster-generating code, nor on 3D formats. The applicability of our findings to these broader settings is unknown.

(f) Negative control scope. Our Python negative control (§4.6) shows that non-visual code yields 69–70% of the visual cross-format CKA in instruction-tuned models, indicating that shared concept or prompt structure—rather than visual semantics alone—contributes substantially to the measured alignment. However, the Python snippets are randomly sampled (not semantically matched to the visual triples), which limits the precision of this comparison. A stronger control would use semantically matched non-visual code triples (e.g., Python programs describing analogous computational structures), which remains a non-trivial dataset construction challenge.

References

Yonatan Belinkov. 2022. Probing classifiers: Promises, shortcomings, and advances. *Computational Linguistics*, 48(1):207–219.

Jonas Belouadi and Steffen Eger. 2024. Automatikz: Text-guided synthesis of scientific vector graphics with tikz. In *Proceedings of ICLR*.

Jonas Belouadi, Foteini Simistira Liwicki, and Steffen Eger. 2024. Datikzv2: Scaling up the generation of scientific figures with tikz. In *Findings of EMNLP*.

Alexis Conneau, German Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni. 2018. What you can cram into a single $&!#* vector: Probing sentence embeddings for linguistic properties. In *Proceedings of ACL*.

Frances Ding, Jean-Stanislas Denain, and Jacob Steinhardt. 2021. Grounding representation similarity through statistical testing. In *Advances in Neural Information Processing Systems*, volume 34.

John Hewitt and Christopher D. Manning. 2019. A structural probe for finding syntax in word representations. In *Proceedings of NAACL-HLT*.

Binyuan Hui, Jian Yang, Zeyu Cui, Jiaxi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, and 1 others. 2024. Qwen2.5-coder technical report. *arXiv preprint arXiv:2409.12186*.

Ganesh Jawahar, Benoît Sagot, and Djamé Seddah. 2019. What does BERT learn about the structure of language? In *Proceedings of ACL*.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. 2019. Similarity of neural network representations revisited. In *Proceedings of ICML*.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of SOSP*.

Ari S. Morcos, Maithra Raghu, and Samy Bengio. 2018. Insights on representational similarity in neural networks with canonical correlation. *Advances in Neural Information Processing Systems*, 31.

Thao Nguyen, Maithra Raghu, and Simon Kornblith. 2022. Do wide neural networks really need to be wide? A reprogramming perspective on CKA limitations. In *Advances in Neural Information Processing Systems*, volume 35.

Yuansheng Ni, Songcheng Cai, Xiangchao Chen, Jiarong Liang, Zhiheng Lyu, Jiaqi Deng, Kai Zou, Ping Nie, Fei Yuan, Xiang Yue, and Wenhu Chen. 2025. Viscoder2: Building multi-language visualization coding agents. *arXiv preprint arXiv:2510.23642*.

Kiho Park, Yo Joong Choe, and Victor Veitch. 2024. The linear representation hypothesis and the geometry of large language models. *arXiv preprint arXiv:2311.03658*.

Telmo Pires, Eva Schlinger, and Dan Garrette. 2019. How multilingual is multilingual BERT? In *Proceedings of ACL*.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, and 1 others. 2021. Learning transferable visual models from natural language supervision. In *Proceedings of ICML*.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of EMNLP*.

Charly Reux, Mathieu Acher, Djamel Eddine Khelladi, Olivier Barais, and Clément Quinton. 2025. Llm code customization with visual results: A benchmark on tikz. In *Proceedings of EASE*.

Kriz Tahimic and Charibeth Cheng. 2025. Mechanistic interpretability of code correctness in llms via sparse autoencoders. *arXiv preprint arXiv:2510.02917*.

Shuai Tang, Wesley J. Maddox, Charlie Dickens, Tom Diethe, and Andreas Damianou. 2020. Similarity of neural network models: A survey of functional and representational measures. *arXiv preprint arXiv:2012.14421*.

Ronghuan Wu, Wanchao Su, and Jing Liao. 2025. Chat2svg: Vector graphics generation with large language models and image diffusion models. In *Proceedings of CVPR*.

Shijie Wu and Mark Dredze. 2019. Beto, bentz, becas: The surprising cross-lingual effectiveness of BERT. In *Proceedings of EMNLP*.

Xiaoming Xing. 2024. SVGX-Core-250k: A large-scale dataset of captioned SVG icons. HuggingFace: xingxm/SVGX-Core-250k.

Xiaoming Xing. 2025. VisPlotBench: A benchmark for visual code generation across formats. Used for eval pool construction (SVG, TikZ, Asymptote test splits).

Xiaoming Xing and 1 others. 2025a. VisCode-Multi-679K: Multi-language visualization SFT corpus. Training data for VisCoder2; includes VCM-SVG, VCM-TikZ, VCM-Asy subsets.

Ximing Xing, Juncheng Hu, Guotao Liang, Jing Zhang, Dong Xu, and Qian Yu. 2025b. Empowering llms to understand and generate complex vector graphics. In *Proceedings of CVPR*.

Zhe Yin, Xiaodong Gu, and Beijun Shen. 2025. Neuron-guided interpretation of code llms: Where, why, and how? *arXiv preprint arXiv:2512.19980*.

Bocheng Zou, Mu Cai, Jianrui Zhang, and Yong Jae Lee. 2024. Vgbench: Evaluating large language models on vector graphics understanding and generation. In *Proceedings of EMNLP*.

Appendix

Eval pool construction. The evaluation pool comprises 200 captioned visual programs per format (600 total) from held-out test splits: SVG from VisPlotBench (Xing, 2025) and VCM-SVG (Xing et al., 2025a); TikZ from DaTikZ

Table A1: Complete CKA similarity values per model, layer, and format pair. Mean is the arithmetic average of the three format pairs, consistent with main text Table 1.

Model	Layer	SVG–TikZ	SVG–Asy	TikZ–Asy	Mean
Qwen2.5-Coder	L4	0.049	0.092	0.120	0.087
Qwen2.5-Coder	L8	0.059	0.087	0.125	0.090
Qwen2.5-Coder	L12	0.067	0.098	0.200	0.121
Qwen2.5-Coder	L16	0.086	0.124	0.212	0.141
Qwen2.5-Coder	L20	0.087	0.118	0.180	0.128
Qwen2.5-Coder	L24	0.105	0.109	0.209	0.141
Qwen2.5-Coder	L28	0.103	0.114	0.229	0.149
VisCoder2	L4	0.050	0.092	0.122	0.088
VisCoder2	L8	0.062	0.089	0.133	0.095
VisCoder2	L12	0.075	0.103	0.218	0.132
VisCoder2	L16	0.089	0.124	0.216	0.143
VisCoder2	L20	0.086	0.124	0.181	0.130
VisCoder2	L24	0.102	0.125	0.208	0.145
VisCoder2	L28	0.093	0.116	0.203	0.137
Qwen2.5-Base	L4	0.050	0.087	0.118	0.085
Qwen2.5-Base	L8	0.058	0.087	0.122	0.089
Qwen2.5-Base	L12	0.061	0.093	0.182	0.112
Qwen2.5-Base	L16	0.067	0.112	0.166	0.115
Qwen2.5-Base	L20	0.070	0.104	0.153	0.109
Qwen2.5-Base	L24	0.086	0.099	0.177	0.121
Qwen2.5-Base	L28	0.044	0.081	0.104	0.076

Algorithm 1 A2 Permutation Null Test for Cross-Format CKA

Require: Hidden-state matrices $\mathbf{H}^{(m,f,l)} \in \mathbb{R}^{N\times d}$ for model m, format f, layer l
Require: Number of permutations $B = 5{,}000$
$\mathbf{H}_0$**:** Cross-format CKA arises from chance alignment of independently sampled representations.
for each model m **do**
  Compute observed CKA for each (l, f_1, f_2) pair from aligned Gram matrices
  $\bar{c}_{\text{obs}} \leftarrow$ mean over all layers and pairs
  for $b = 1$ to B **do**
    Draw random permutation π of $\{1, \ldots, N\}$
    for each layer l and pair (f_1, f_2) **do**
      Permute rows and columns of $\mathbf{K}^{(f_2)}$ by π: $\tilde{\mathbf{K}}^{(f_2)}_{ij} = \mathbf{K}^{(f_2)}_{\pi(i),\pi(j)}$
      Compute CKA$(\mathbf{K}^{(f_1)}, \tilde{\mathbf{K}}^{(f_2)})$
    end for
    $\bar{c}^{(b)}_{\text{null}} \leftarrow$ mean over all layers and pairs
  end for
  $p \leftarrow \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}[\bar{c}^{(b)}_{\text{null}} \geq \bar{c}_{\text{obs}}]$
end for
Ensure: Per-model p-value, null distribution $\{\bar{c}^{(b)}_{\text{null}}\}_{b=1}^{B}$, observed/null ratio

v1 test (Belouadi and Eger, 2024); Asymptote from VisPlotBench and VCM-Asy (Xing et al., 2025a). All sources are filtered uniformly: code length $\in [50, 8000]$ characters, caption length $\in [10, 3000]$ characters, render success within 30s, and normalized-code-hash deduplication.

Generation protocol. Each eval caption is presented to each model under two shot regimes: **0-shot** (caption only) and **3-shot** (three in-context learning exemplars from independent training data, disjoint from both eval and probe pools). This yields 200 $\times$ 3 formats $\times$ 2 regimes $\times$ 3 models $=$ 3,600 free generations. Generation uses vLLM (Kwon et al., 2023) in fp16 with temperature T=0.3, top_p=1.0, n=1, and format-specific max tokens (SVG/TikZ: 1024, Asy: 2048) and stop tokens.

Hybrid evaluation. Content fidelity uses a hybrid metric: visual attributes with CLIPScore (Radford et al., 2021) SNR $\geq$ 1.0 (color, shape count, stroke/fill) are scored via CLIPScore (ViT-L/14); attributes with SNR $<$ 1.0 (quadrant, spatial relation) are scored via programmatic attribute-match from the code AST, addressing CLIP's known weakness in spatial semantics. The SNR gate was calibrated in a pilot study (n=20, 5 dimensions) using a shuffle-permutation null.

Table A2: Procrustes distance per model, layer, and format pair. Lower values indicate greater representational similarity after optimal rotation.

Model	**Layer**	**SVG–TikZ**	**SVG–Asy**	**TikZ–Asy**	**Mean**
Qwen2.5-Coder	L4	0.073	0.095	0.139	0.102
Qwen2.5-Coder	L8	0.079	0.095	0.146	0.107
Qwen2.5-Coder	L12	0.090	0.106	0.217	0.138
Qwen2.5-Coder	L16	0.113	0.134	0.226	0.158
Qwen2.5-Coder	L20	0.113	0.128	0.198	0.146
Qwen2.5-Coder	L24	0.124	0.128	0.219	0.157
Qwen2.5-Coder	L28	0.117	0.124	0.220	0.153
VisCoder2	L4	0.072	0.094	0.141	0.102
VisCoder2	L8	0.080	0.095	0.150	0.108
VisCoder2	L12	0.095	0.108	0.222	0.142
VisCoder2	L16	0.115	0.133	0.222	0.157
VisCoder2	L20	0.115	0.133	0.195	0.148
VisCoder2	L24	0.123	0.136	0.216	0.158
VisCoder2	L28	0.110	0.125	0.202	0.146
Qwen2.5-Base	L4	0.071	0.093	0.136	0.100
Qwen2.5-Base	L8	0.077	0.091	0.141	0.103
Qwen2.5-Base	L12	0.083	0.097	0.201	0.127
Qwen2.5-Base	L16	0.093	0.113	0.193	0.133
Qwen2.5-Base	L20	0.096	0.112	0.177	0.128
Qwen2.5-Base	L24	0.104	0.112	0.195	0.137
Qwen2.5-Base	L28	0.065	0.084	0.133	0.094

Table A3: Projection-Weighted CCA (PWCCA; Morcos et al. 2018) mean similarity per model and layer. Higher values indicate stronger cross-format alignment.

Model	**L4**	**L8**	**L12**	**L16**	**L20**	**L24**	**L28**
Qwen2.5-Coder	0.770	0.773	0.813	0.819	0.799	0.779	0.779
VisCoder2	0.768	0.774	0.805	0.809	0.789	0.776	0.771
Qwen2.5-Base	0.759	0.766	0.801	0.807	0.793	0.776	0.811

Table A4: Format-classification probe accuracy with real vs. permuted labels. All 21 model–layer cells show identical values: real accuracy 1.000, permuted 0.333, gap 0.667. This confirms that probe accuracy reflects genuine format information rather than memorized structure.

Model	**L4**	**L8**	**L12**	**L16**	**L20**	**L24**	**L28**
Real accuracy							
Qwen2.5-Coder	1.000	1.000	1.000	1.000	1.000	1.000	1.000
VisCoder2	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Qwen2.5-Base	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Permuted-label accuracy (100 permutations)							
All models	0.333	0.333	0.333	0.333	0.333	0.333	0.333

Table A5: A2 permutation test (n=5,000): observed cross-format CKA mean vs. null distribution obtained by permuting sample identities within each format. $p<0.001$ for all models confirms that cross-format representational similarity is not an artifact of format-level statistics.

Model	**Obs. CKA Mean**	**Null Mean**	**Null Std**	p**-value**
Qwen2.5-Coder	0.123	0.041	0.003	<0.001
VisCoder2	0.124	0.044	0.003	<0.001
Qwen2.5-Base	0.101	0.037	0.003	<0.001

Table A6: Format classification probe accuracy (5-fold CV, logistic regression). All 21 model–layer cells are 1.0000.

Model	L4	L8	L12	L16	L20	L24	L28
Qwen2.5-Coder	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
VisCoder2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Qwen2.5-Base	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table A7: PWCCA permutation null calibration (300 perms). The observed/null gap is statistically significant but small (ratio $\sim$1.1$\times$), compared to CKA's $\sim$3$\times$ null ratio.

Model	Observed	Null (mean±std)	Ratio	p
Qwen2.5-Coder	0.790	0.688±0.004	1.148×	<0.001
VisCoder2	0.785	0.685±0.004	1.145×	<0.001
Qwen2.5-Base	0.788	0.699±0.004	1.127×	<0.001

A1 power simulation data-generating process. The A1 power simulation operates at a single representative layer (L24, chosen as the layer with the highest mean CKA across clean models). In each of 1,000 Monte Carlo iterations, we resample n=252 triples with replacement from the original probe pool, recompute Gram matrices for each model–format combination, and calculate the mean CKA across 2 models $\times$ 3 format pairs (N=6 headline cells) or 3 models $\times$ 3 pairs (N=9 auxiliary cells). The null threshold is the 95th percentile of a permutation null distribution (500 permutations of sample indices). Power is the pro-

Table A8: SBERT threshold sensitivity: mean CKA at L28 by matching threshold. All 252 triples have min cosine ≥ 0.70, so thresholds 0.60 and 0.65 yield identical results (N=252). At thresholds ≥ 0.75, N drops sharply, making CKA estimates unreliable due to $n \ll d$ inflation (Nguyen et al., 2022). The relative ordering TikZ–Asy > SVG–Asy > SVG–TikZ is preserved at N=51, consistent with the syntax-family confound (F3). Main results are insensitive to threshold choice in the 0.60–0.70 range.

Threshold	N	Coder	VisCoder2	Base
0.60	252	0.149	0.137	0.076
0.65	252	0.149	0.137	0.076
0.70	252	0.149	0.137	0.076
0.75	51	0.252	0.247	0.146
0.80	6	0.739	0.766	0.616

Table A9: Stage A generation summary: validity rate per model and format (3,600 total, 0-shot and 3-shot pooled). Temperature T=0.3, n=1 per prompt. (Table A9)

Model	SVG	TikZ	Asy
Qwen2.5-Coder	1.000	0.995	1.000
VisCoder2	0.972	0.962	1.000
Qwen2.5-Base	0.662	0.722	0.752
Overall	3,227/3,600 valid (89.6%)		

portion of iterations where the bootstrapped mean CKA exceeds the null threshold; power $= 1.0$ at both N=6 and N=9 confirms that the permutation test can detect signals of the observed magnitude.

PWCCA in the $n \ll d$ regime. The limited discriminative power of PWCCA in this study reflects known properties of CCA when sample size is much smaller than feature dimensionality. With n=252 samples and d=3,584 features (ratio 0.07), canonical correlations converge toward 1.0 regardless of true signal strength, compressing the dynamic range available to distinguish signal from noise. This is the same degeneracy that motivated our removal of raw CCA from the analysis pipeline. PWCCA corroborates the *direction* of the CKA signal—observed values exceed the permutation null for all three models—but its observed/null ratio (1.13–1.15$\times$) is far below CKA's ($\sim$3$\times$), limiting its value as an independent robustness check in this experimental design.